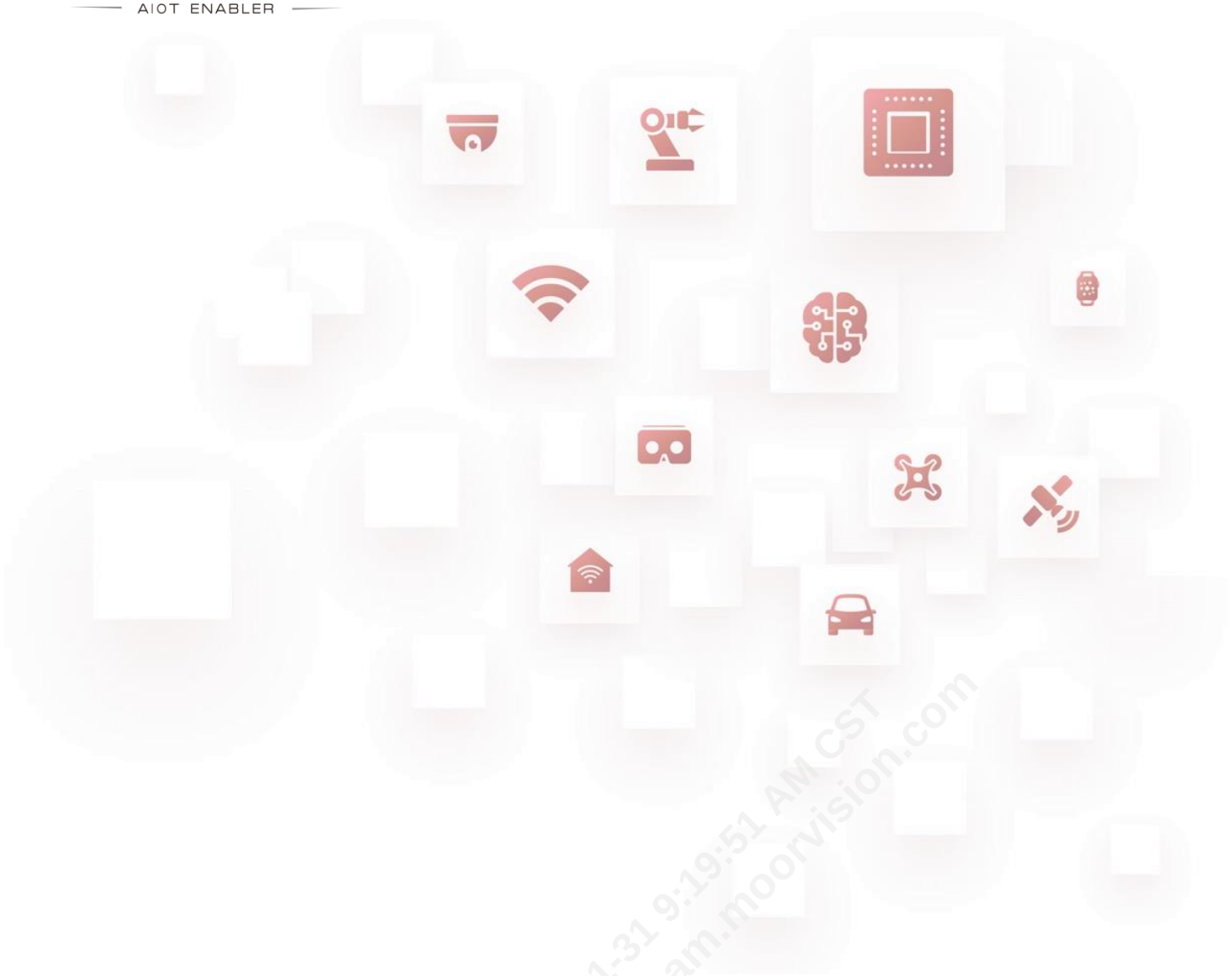
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Thundercomm TurboX SOM SDK Git

User Guide

Rev. V1.1
May 12, 2023

DN: tc-A-163211-en

Revision History

Revision	Date	Author	Description
1.0	Apr 26, 2023	Lining Jiao	Initial release.
1.1	May 12, 2023	Lining Jiao	Add Chapter 8. Create Token.

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About This Document

- This document is applicable to all SOM modules.
- Illustrations in this documentation might look different from your product.
- Depending on the model, some optional accessories, features, and software programs might not be available on your device.
- Depending on the version of operating systems and programs, some user interface instructions might not be applicable to your device.
- Documentation content is subject to change without notice. Thundercomm makes constant improvements on the documentation of the products, including this guidebook.
- Function declarations, function names, type declarations, attributes, and code samples appear in a different format, for example, `cp armcc armcpp`.
- Code variables appear in angle brackets, for example, `<number>`.
- Button, tool, and key names appear in bold font, for example, click **Save** or press **Enter**.
- Commands to be entered appear in a different font; on the host computer use $ as shell prompt, while on the target device use # as shell prompt, for example,

```
$ adb devices
# logcat
```

- Part of the code that does not contain instructions appear in a different format, for example,

```
SUBSYSTEM=="usb", ATTR{idVendor}=="18d1", MODE="0777", GROUP="adm"
```

- Folders, path and files are formatted in italic, for example, *turbox_flash_flat.sh*.

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Chapter 1. Overview

sdkgit.thundercomm.com is a GitLab-based system developed by Thundercomm for releasing SOM SDK (hereinafter referred to as "**SDK Git**").

This document introduces the operations that you can perform with SDK Git, including but not limited to the following:

- System login
- View SDK code
- Download SDK code
- Upload source code

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Chapter 2. System Login

Address: https://sdkgit.thundercomm.com

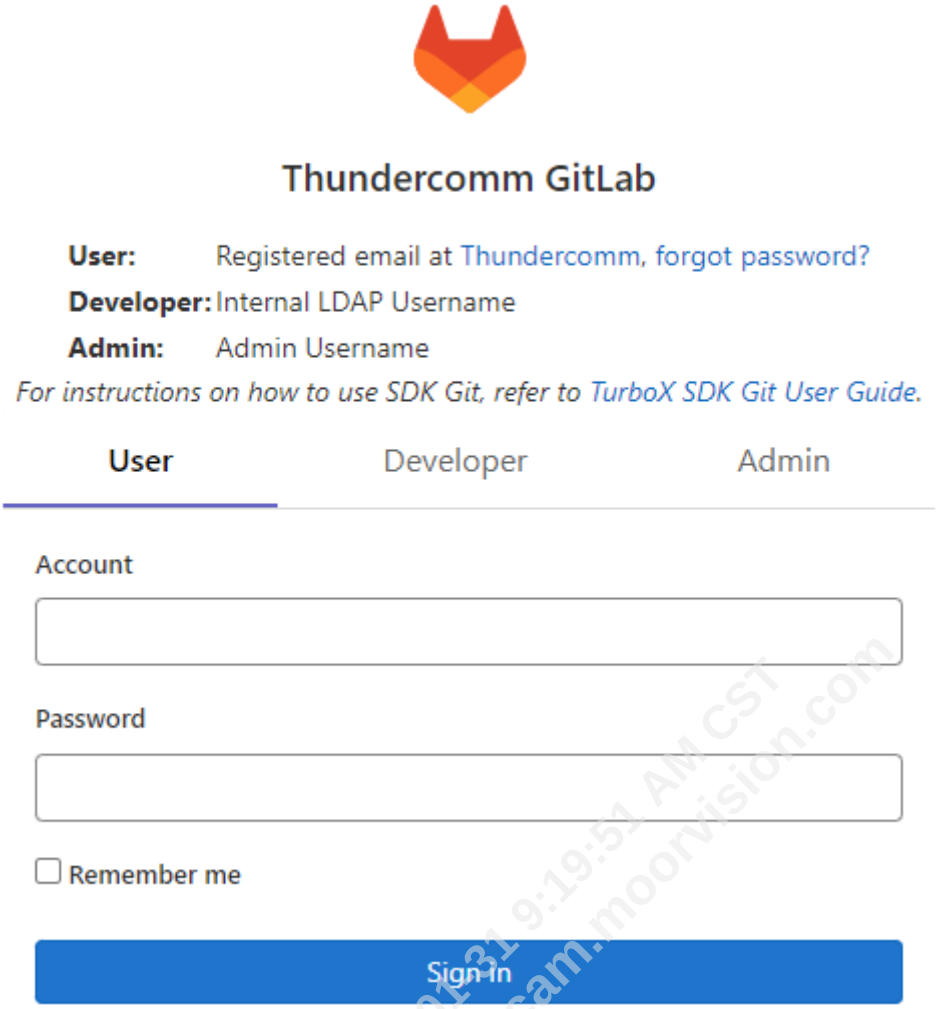


Figure 2-1. System Login Screen

- **Login interface for customers**
  - Login as: **User**
  - Account: Registered email on Thundercomm website
  - Password: User password on Thundercomm website
- **Login interface for internal staff**
  - Login as: **Developer**
  - Account: Internal LDAP username
  - Password: Internal LDAP password

Chapter 3. Configure SSH Public Key

After logging in to the system, it is necessary to configure SSH public key before SDK download or code upload.

Step 1. Generate SSH public key locally.

Execute `ssh-keygen -t rsa -C "your email address"` in terminal. Then the SSH public key will be saved to the *~/.ssh/id_rsa.pub* file.

```
::~$ ssh-keygen -t rsa -C "
Generating public/private rsa key pair.
Enter file in which to save the key (/home/ /.ssh/id_rsa):
Created directory '/home/ /.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ /.ssh/id_rsa
Your public key has been saved in /home/ /.ssh/id_rsa.pub
The key fingerprint is:
SHA256:a9kV12 Lmrd2x7RmQR0phvgk
The key's randomart image is:
+---[RSA 3072]----+
|             o +*|
|            o=o+*|
|          +E*++.=|
|         + oo*O.+|
|        S . =*.O.|
|         + +..*++|
|        + o .+oo.|
|       .    .  .o|
|               ..|
+----[SHA256]-----+
```

Figure 3-1. Log Output of SSH Public Key Generation (Example on Ubuntu)

Step 2. Upload the SSH public key to SDK Git.

1) Execute `cat ~/.ssh/id_rsa.pub` in terminal to obtain and copy the SSH public key.

```
:~$ cat ~/.ssh/id_rsa.pub
ssh-rsa AAAAB3NzaC1yc                                XGOXjB1ZH5oVq6syXOpH5/nHljP4
VDBemL8834B1ZrRcMg2wF                                IbAFRBdmRsyRqct/keGJkz6HRNh8
LEvCEPO0X4U/M9L1KKuvC                                Tn/qT946Fx+QsoixiqC4XaVpFJca
9sd0m+nChu16yGxJt31J6                               P4LbgcVQCHbS3MH5gLmS0Suy4T/R
N/paIHWVadb/hGWwNUO+v                                dgvxv5wSd38BLfCoCyq+v3WByo+B
LOQh28fxmcB4yeTqr6NOM                               QPXA+LkFZaUsIuh6uinUNBFAtZR4
Z0uZLHG/qAU8MD1zTHkA8                               jCTd4oGSW0QGU5aqtl8bGzsr70=
```

Figure 3-2. Screen for Obtaining SSH Public Key

2) Upload the SSH public key to SDK Git.

Go to https://sdkgit.thundercomm.com/-/profile/keys, paste the SSH public key and then submit it with the **Add Key** button.

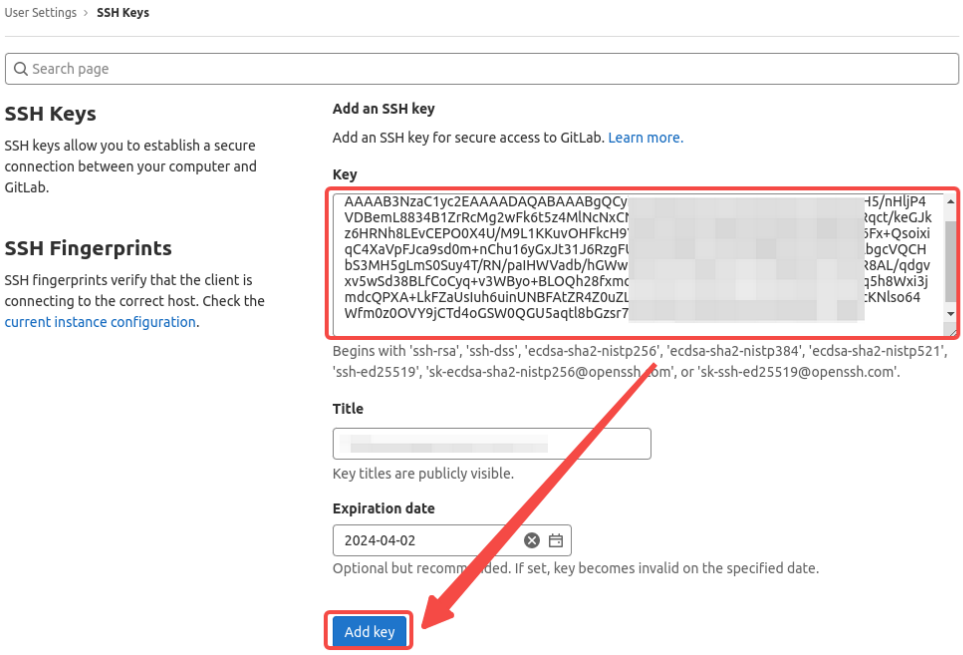


Figure 3-3. Upload SSH Public Key

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Chapter 4. Group Classification

Upon login to the system, you can view the authorized groups by clicking "groups" in the top navigation bar:

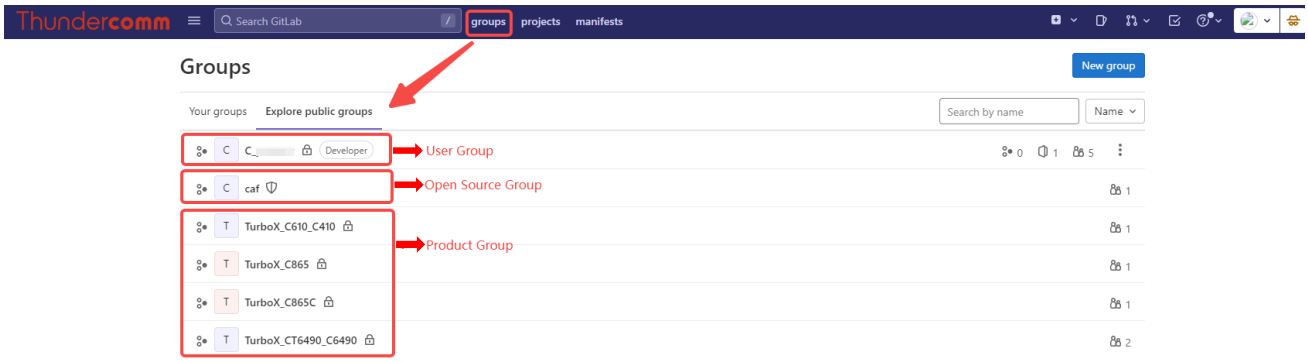


Figure 4-1. Group Example

Groups in SDK Git are divided into three categories:

- **TurboX_<Product Name>:** Product Group, used to manage product private codes and open source modified by Thundercomm; customers can only view the products to which they have access. In this group, customers are taken as “reporter”, who can browse and download codes, but cannot submit codes.
- **caf:** Open Source Group, used to manage open sources that have not been modified by Thundercomm; all products share a common code library. In this group, customers are taken as “reporter”, who can browse and download codes, but cannot submit codes.
- **C_< user's email suffix >:** User Group, used to manage customers' members and private codes. In this group, customers are taken as “developer”, who can browse, download and upload codes.

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Chapter 5. Code Download

5.1. Full SDK code download

➲ **NOTE:** Refer to the release notes of each SDK version.

Proceed with the steps below to obtain SDK release notes in the manifest project of each product.

Step 1. Click "manifests" in the top navigation bar to view the authorized product manifest project.

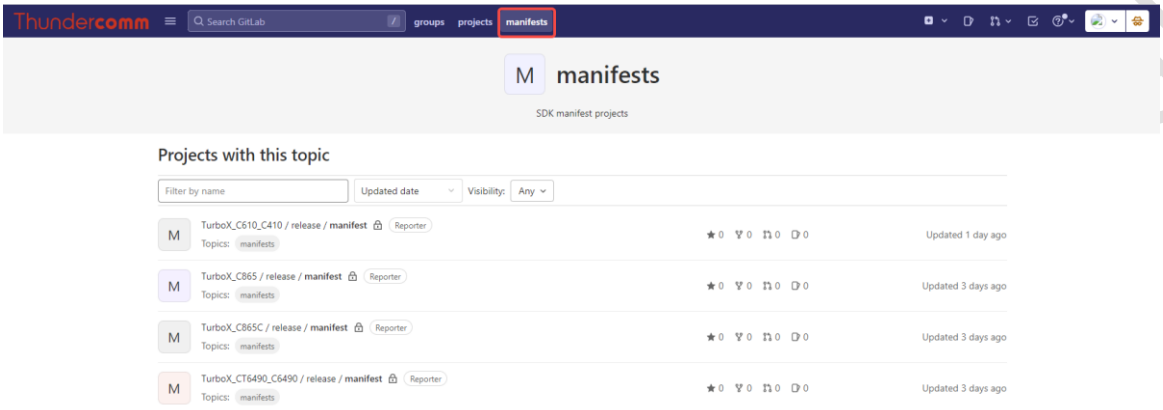


Figure 5-1. Screen for Obtaining Manifest Project

Step 2. Enter the manifest project, and you can find the release notes of each SDK version on the "Deployments -> Releases" interface.

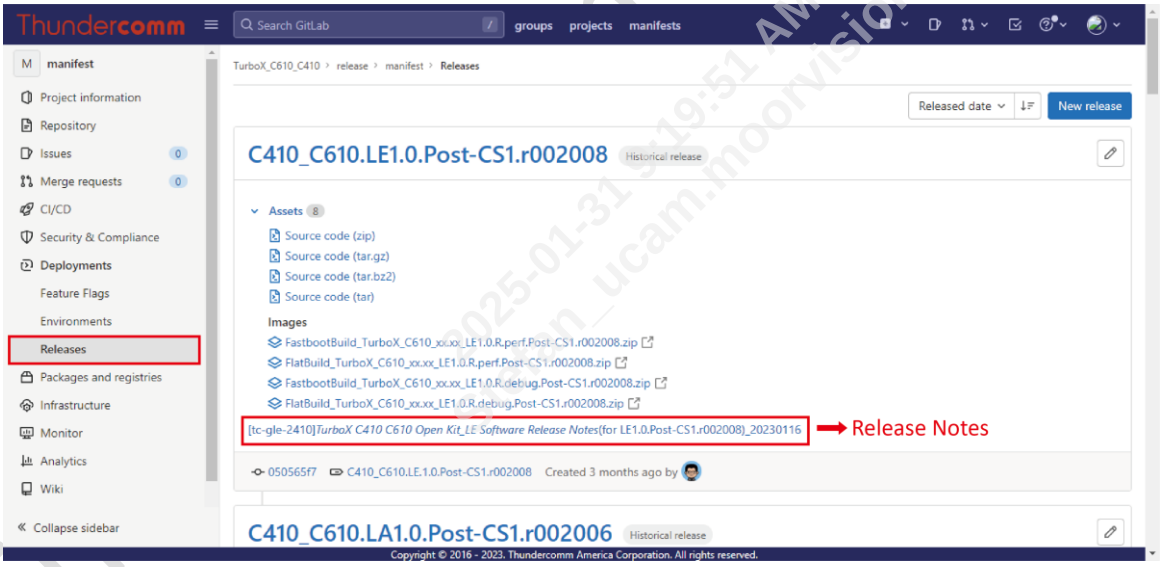


Figure 5-2. Screen for Obtaining Release Notes

5.2. Git repository download

Step 1.Obtain the manifest file of the related SDK version.

Enter the manifest project and select a branch or tag to obtain the corresponding manifest file.

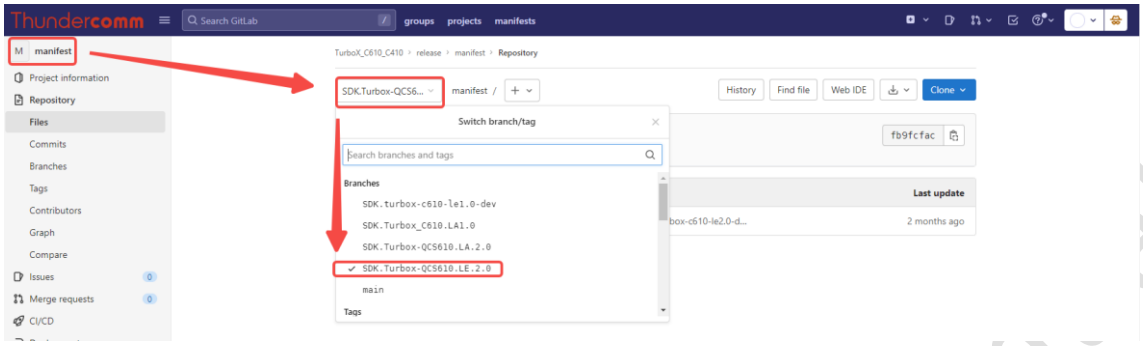


Figure 5-3.Search for Manifest Branch/Tag

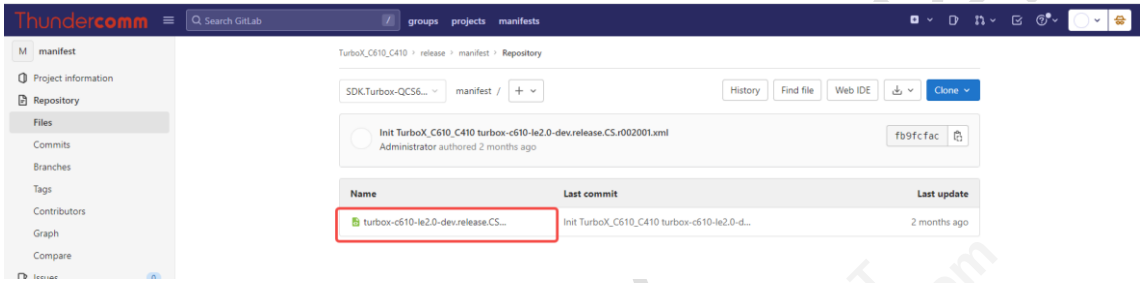


Figure 5-4.Search for Manifest File

Step 2.Obtain the name of the desired Git repository.

Open the manifest file and search for the name of the Git repository.

➲ **NOTE:** The last field of the name in the manifest file indicates the name of the Git repository.

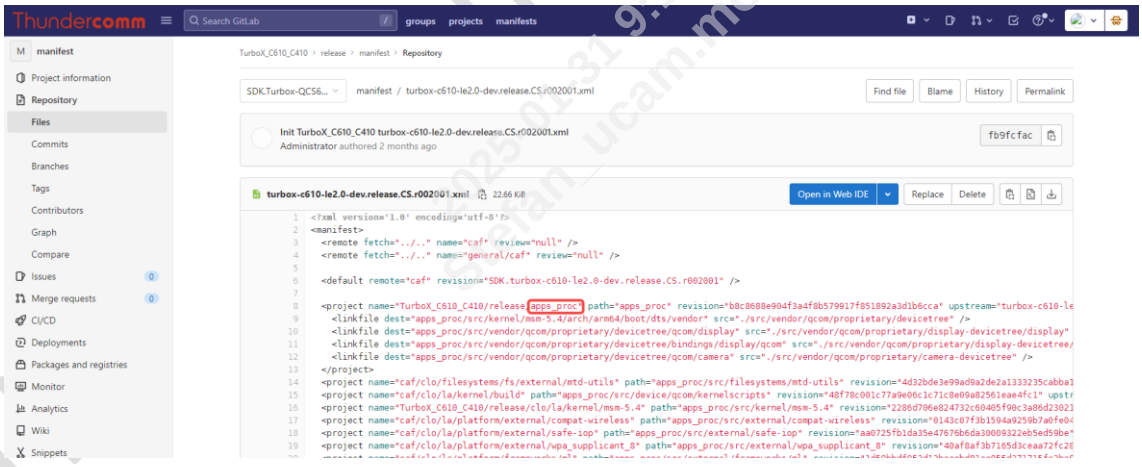


Figure 5-5.Name of Git Repository in Manifest File

Step 3.Access the Git repository.

1) Click "Projects" in the top navigation bar.

2) Search for the name of the Git repository obtained in Step 2.

3) Click the Git repository link to enter the Git repository interface.

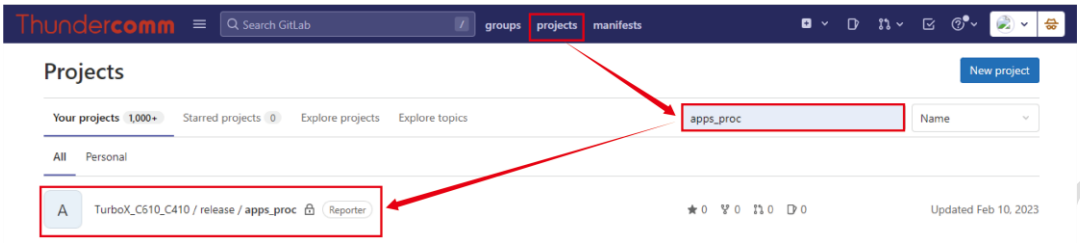


Figure 5-6.Retrieve Git Repository

Step 4.Select the corresponding tag.

Go to the Git repository page and select the desired SDK version tag.

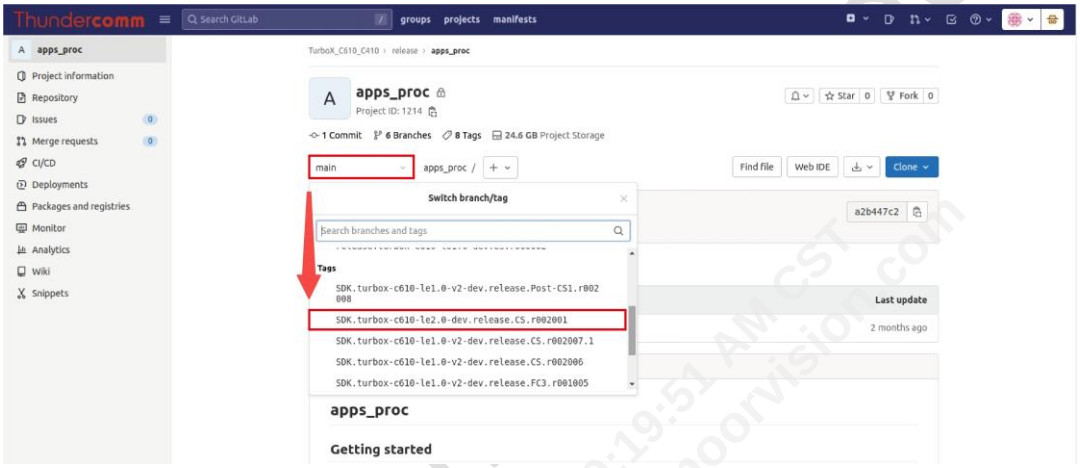


Figure 5-7.Select SDK Version

Step 5.Clone the Git repository.

1) Click "clone" → "Clone with SSH" to copy the command line.

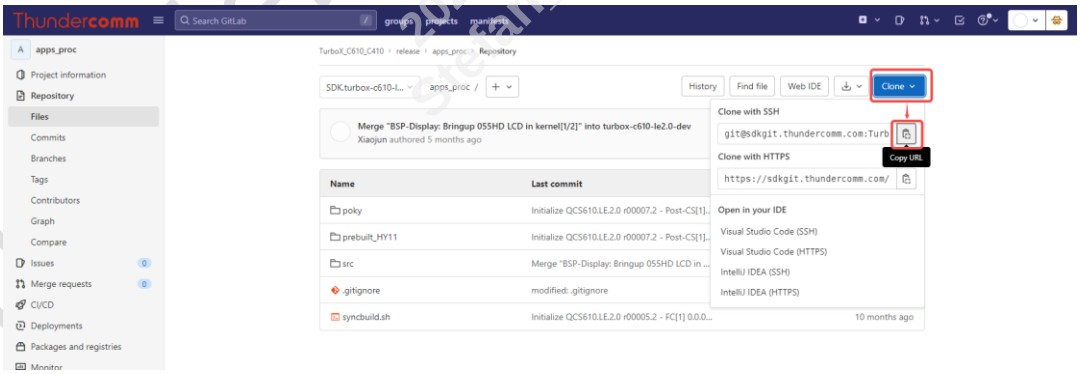


Figure 5-8.Copy Command Line

2) Execute the `Git clone` command locally to download the codes. Parameters following `Git clone` stand for the command line copied in the previous sub-step.

```
[                                    ]$ git clone git@sdkgit.thundercomm.com:TurboX_C610
_C410/release/LINUX/android/vendor/qcom/proprietary.git
Cloning into 'proprietary'...
remote: Enumerating objects: 59679, done.
remote: Total 59679 (delta 0), reused 0 (delta 0), pack-reused 59679
Receiving objects: 100% (59679/59679), 2.11 GiB | 1.35 MiB/s, done.
Resolving deltas: 100% (22393/22393), done.
```

Figure 5-9.Download Codes

5.3. File download

Access the to-be-downloaded file on the Git repository page, and click the download button.

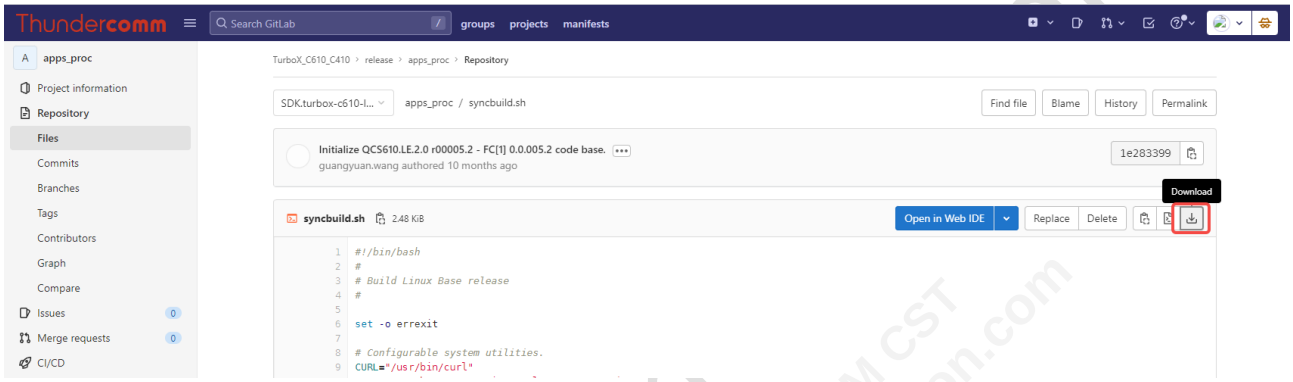


Figure 5-10. Download Specified File

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Chapter 6. View Code Differences of Various Versions

For code differences of various versions, click "**Repository**" → "**Graph**" on the Git repository page.

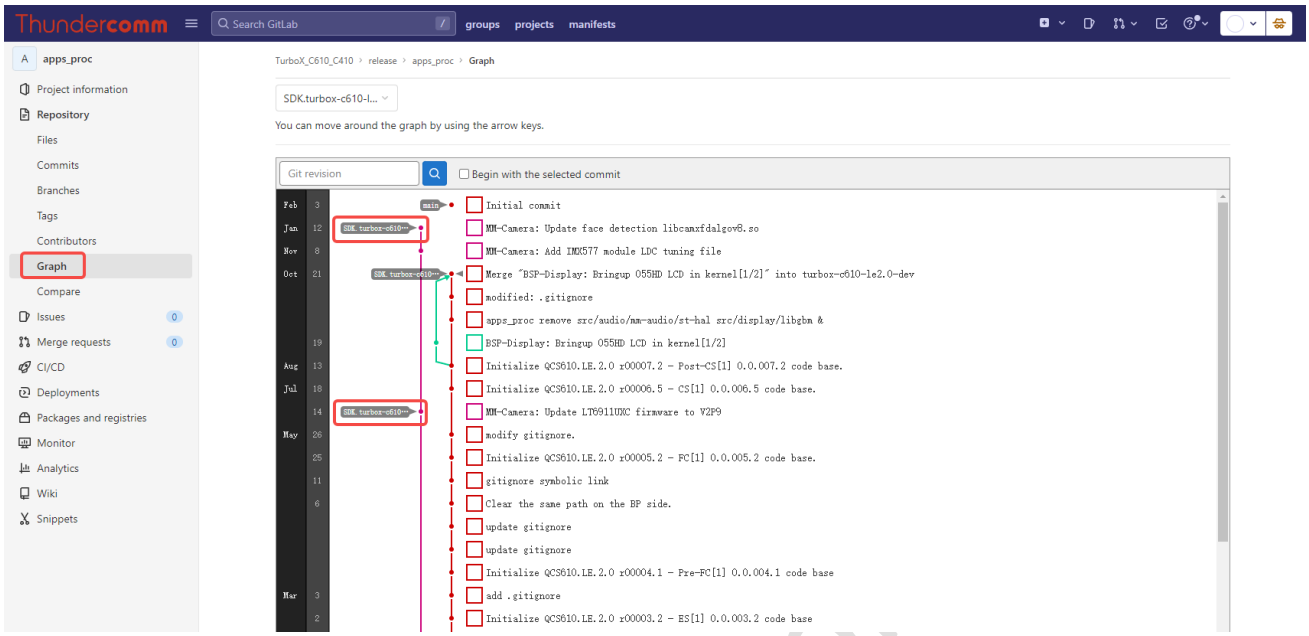


Figure 6-1. View Code Differences

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Chapter 7. Upload Source Code

Follow the steps below to upload codes to your own **User Group**.

Step 1. Create a Git repository.

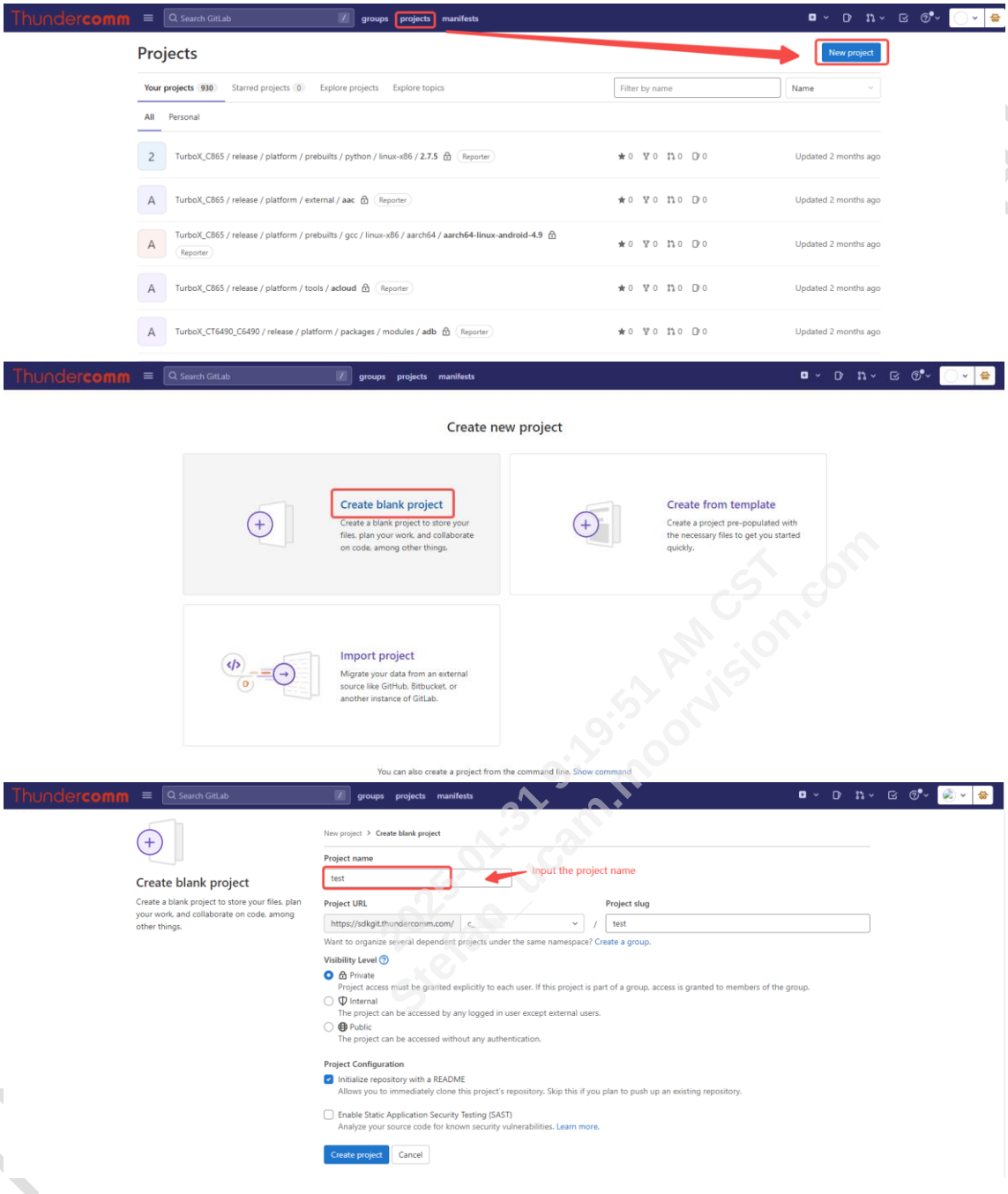


Figure 7-1. Create Git Repository

Step 2. Upload codes.

➲ **NOTE:** Users can upload codes by following the prompts in the Git repository.

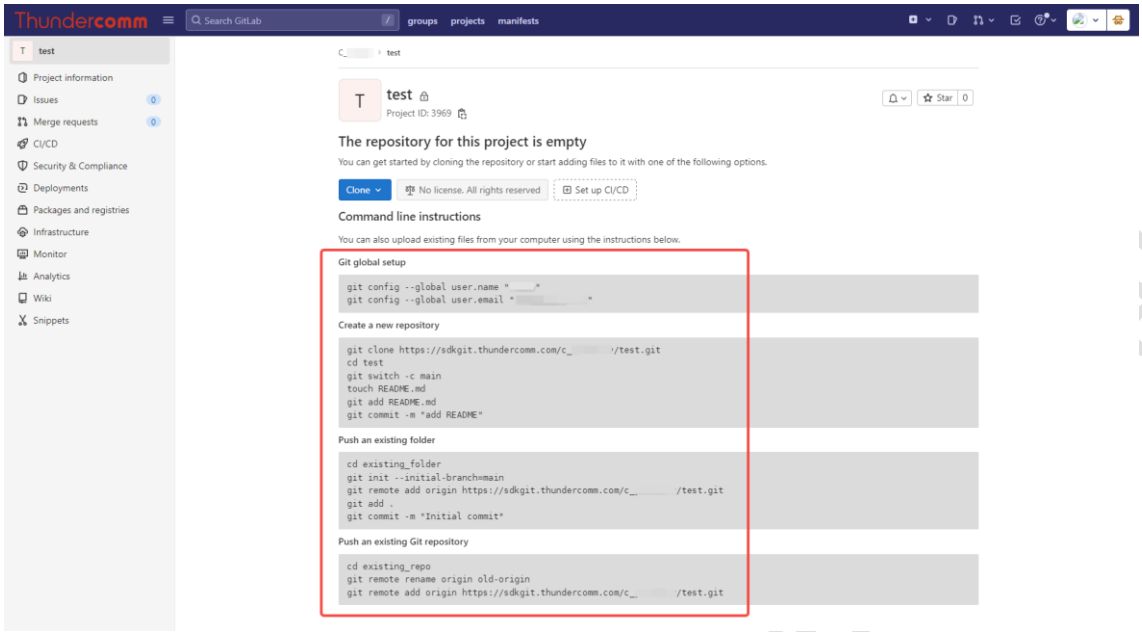


Figure 7-2. Upload Codes

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Chapter 8. Create Token

When initializing Thundercomm TurboX SOM SDK Manager, it is necessary to fill in the token on SDK Git in order to download code from SDK Git. This chapter introduces how to create the token on SDK Git.

Refer to the detailed procedure below.

Step 1. Proceed with the following sub-steps to create the token.

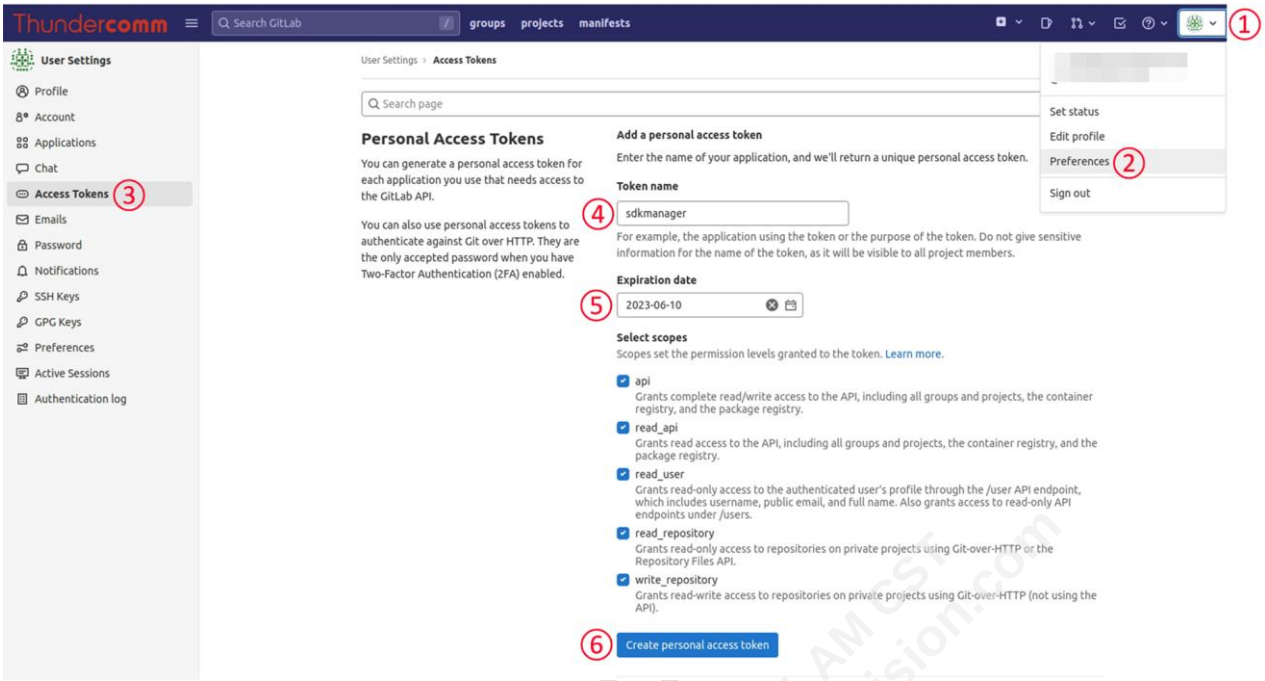


Figure 8-1. Create the Token

1) Click ① → ② → ③ in sequence.

2) Fill in the token name and expiration date in the fields of ④ and ⑤, select all scopes and click ⑥ to complete creation.

Step 2. Click the button as shown in Figure 8-2 to copy the token.

➲ **NOTE:** You can only copy the token on the creation completion interface. If you exit this interface, you will be unable to perform the copy operation.

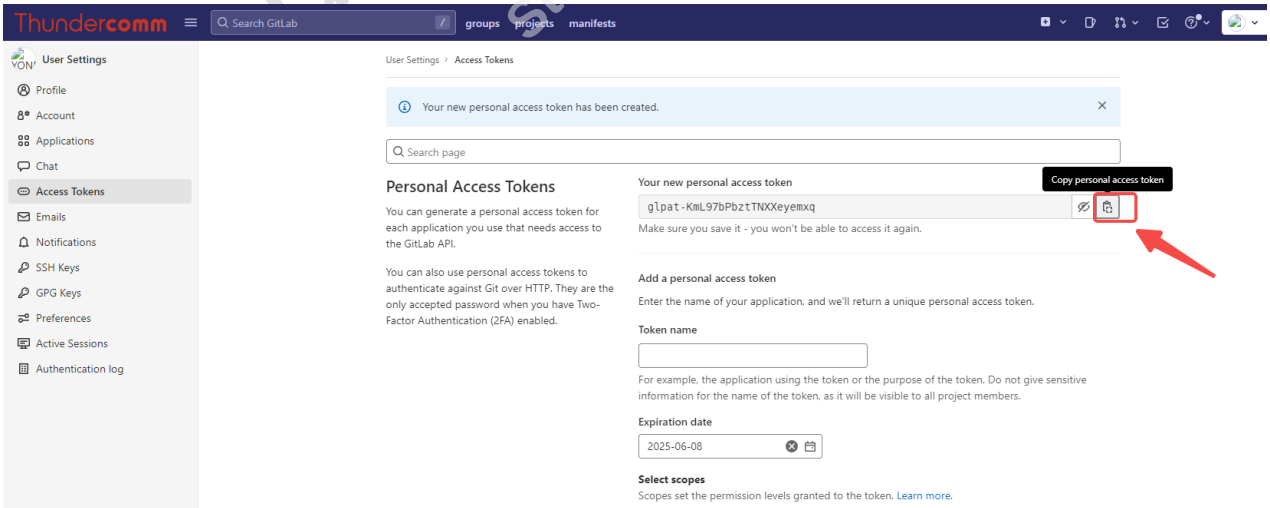


Figure 8-2. Copy the Token

Chapter 9. Miscellaneous Instructions

For more usage of the SDK Git, please refer to: https://sdkgit.thundercomm.com/help.

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